MIL-L-51293B(MU)
10 June 1970
SUPERSEDING
MIL-L-51293A(MU)
29 May 1968

MILITARY SPECIFICATION

LAUNCHER AND GRENADES, SMOKE:

HC AND WP, M176; COMPONENTS FOR

1. SCOPE

1.1 This specification covers the components of the M176 launcher.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-1168 - Lot Numbering of Ammunition.

DRAWINGS

US ARMY MUNITIONS COMMAND

EDGEWOOD ARSENAL

DL13-12-1 - Launcher and Grenades, Smoke: HC and WP, M176.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

FSC 1330

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UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification

(Application for copies of these ratings, rules and regulations should be addressed to Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM Standards

D883-65aT - Tentative Nomenclature Relating to Plastics.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

3. REQUIREMENTS

3.1 Materials. All materials cited on Drawing DL13-12-1 and on the subsidiary drawings shall conform to the specification thereon or to the specific characteristics set forth on the drawings. All plastic material (glass fiber reinforced polypropylene) shall contain 40 ± 5 percent glass fiber by weight.

3.2 Tube burst pressure. The tube (Drawing D13-12-2) shall withstand an internal hydrostatic pressure of 350 ± 25 pounds per square inch gage (psig) at a distance of 2 ± 1/16 inch from the inside surface of the base when tested in accordance with 4.4.4.1.

3.3 Cap leakage. The assembled and sealed cap shall not leak nor become distorted when subjected to an internal air pressure of 5.5 ± 0.5 psig in accordance with 4.4.4.2.

3.4 Cap hardness. The final material condition of the cap body shall be temper 0 with a maximum hardness limit of 40 when tested in accordance with 4.4.4.3.

3.5 Preproduction. Prior to the start of regular production, a preproduction sample of components shall be produced in accordance with this specification for examination and test (see 4.3).

3.6 Cap push-out. The assembled and sealed cap shall require a force of no less than 1,000 pounds (lb) nor greater than 1,600 lb to push out the cap end disc, when tested in accordance with 4.4.4.4.

3.7 Workmanship. The components shall be free from foreign matter and damage (chips, breaks and cracks) that would adversely affect normal assembly and subsequent functioning.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Supplier's responsibility. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.1.2 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 Preproduction inspection.

4.3.1 Sample. A preproduction sample of 5 plastic components plus a minimum total of 30 additional sets of each plastic component (for Government test) shall be produced and selected in equal quantities from each mold cavity, using the same methods, materials, equipment, and processes as will be used in regular production. In addition, a preproduction sample of 5 aluminum caps plus a minimum total of 30 additional caps (for Government test) shall be produced from the same dies, using the same methods, materials, equipment, and processes as will be used in regular production. Each of the additional 30 plastic components and aluminum caps shall be subjected to all the applicable examinations and non-destructive tests.

4.3.2 Inspection procedure.

4.3.2.1 For examination and nondestructive tests. The preproduction sample shall be examined in accordance with the list of defects and tested in accordance with 4.4.4.2.

4.3.2.2 For destructive test, (tube pressure, cap hardness, and cap push-out). The preproduction sample shall be tested in accordance with 4.4.4.1, 4.4.4.3 and 4.4.4.4 as applicable; these tubes shall not be used for production purposes.

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4.3.3 Acceptance/rejection criteria. The preproduction sample shall comply with the examinations and tests specified herein to be acceptable. The supplier shall obtain written approval from the contracting officer prior to proceeding with regular production.

4.4 Quality conformance inspection.

4.4.1 Lotting. Each lot shall be identified and controlled in accordance with MIL-STD-1168. Contained in anyone interfix number of launchers, no more than one interfix number from one manufacturer of the following is permitted:

(a) One batch (see 6.2) of raw material.

4.4.2 Sampling.

4.4.2.1 For examination. At the beginning and at the completion of production, one or more components shall be taken from each die producing the end item, and from each mold cavity for examination. In the event of a failure or repair to the mold, one or more components shall be taken to requalify the mold. During production, the quantity and frequency of these examinations shall be at the discretion of the supplier, but shall be no less than one component from each mold cavity in every six hours of production.

4.4.2.2 For destructive tests (tube pressure, cap hardness, and cap push-out). Sampling shall be conducted in accordance with MIL-STD-105, level S-4, using an AQL of 2.5 for each test.

4.4.3 Inspection procedure.

4.4.3.1 For examination. Sample components shall be examined in accordance with the list of defects. If one or more than one component fails to comply with the applicable requirement, the components represented shall be rejected back to the last acceptable examination.

4.4.3.2 For tests.

(a) Nondestructive (cap leakage). Each cap in the lot shall be tested in accordance with 4.4.4.2.

(b) Destructive (tube pressure, cap hardness and cap push-out). Sample caps and tubes shall be tested in accordance with 4.4.4.1, 4.4.4.3, and 4.4.4.4 and MIL-STD-105, using an AQL of 2.5 percent defective for each test.

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4.4.3.3 List of defects.

(a) Tube (Dwg D13-12-2).

*1. Cracks
2. Thread missing or incorrect
3. Eccentricity between O.D. of boss and pitch diameter of thread excessive
4. Eccentricity between O.D. and pitch diameter of thread excessive
5. Wall thickness (2 places) minimum
6. Flatness plane on I.D. excessive
7. Taper on I.D. incorrect
8. Inside depth of tube incorrect
9. Cartridge hole diameter incorrect
10. Groove incorrect or incorrectly located
11. Dimension 0.359 incorrect
12. Workmanship (3.7)

*Defined in ASTM designation D883-65aT.

(b) Holder (Dwg D13-12-3).

1. Grenade cradle I.D. incorrect
2. Length of grenade cradle incorrect
3. Locating hole spacing incorrect
4. Overall length incorrect
5. Eccentricity among O.D., undercut and extreme diameter of cavity section incorrect
6. Workmanship (3.7)

(c) Cover (Dwg D13-12-4).

1. I.D. incorrect
2. Locating pin spacing incorrect
3. Overall length incorrect
4. Cutout location incorrect
5. Locating tab incorrect
6. Workmanship (3.7)

(d) Spacer (Dwg B13-12-5).

1. O.D. incorrect
2. Thickness incorrect
3. Location of riser incorrect
4. Overall height 0.81 incorrect
5. Workmanship (3.7)

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4.4.3.3 List of defects (continued).

(e) Retainer (Dwg C13-12-6).

1. Thread incorrect
2. Eccentricity among thread, counterbore and hole excessive
3. Counterbore diameter incorrect
4. Counterbore depth incorrect
5. Length incorrect
6. Wrench pocket incorrect
7. Wrench pocket location incorrect
8. Radius missing
9. Workmanship (3.7)

(f) Cap (Dwg C13-12-7).

1. Min I.D. incorrect
2. Max I.D. incorrect
3. I.D. at offset incorrect
4. Max diameter over sealed end incorrect
5. Dimension formed over end (-2) incorrect (0.070) min
6. Dimension 2.00 incorrect
7. Protective coating, and sealer missing or incorrectly applied
8. Width of groove incorrect
9. Diameter of end (-2) incorrect
10. Depth of groove incorrect
11. Height of bead incorrect
12. Formed over end overlapping bead
13. Length of skirt incorrect
14. Temper incorrect
15. Workmanship (3.7)

(g) Cap (Dwg C13-12-17), alternate design.

1. Min I.D. incorrect
2. Max I.D. incorrect
3. I.D. at offset incorrect
4. Dimension 0.156 incorrect
5. Dimension 1.812 incorrect
6. Dimension 0.410 incorrect
7. Angle 45 degree incorrect
8. Protective coating and sealer missing or incorrectly applied
9. Workmanship (3.7)

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4.4.3.3 List of defects (continued).

(h) Seal (Dwg B13-12-10).

1. Width incorrect
2. I.D. incorrect
3. Thickness incorrect

4.4.4 Tests.

4.4.4.1 Tube pressure. The tube shall be placed in a test fixture so that the tube is sealed at a distance of 2 + 1/16 inches from the inside surface of the base. The internal section of the tube shall then be subjected to the specified pressure at a uniform rate within a period of 5 seconds. Test shall be conducted at ambient temperature. These tubes shall be identified as test items and shall not be used for production purposes.

4.4.4.2 Cap leakage. The assembled and sealed cap shall be placed in a test fixture (see 6.3) and subjected to the specified pressure for a period of 15 to 20 seconds. Visual observation shall start when the gage reaches the required pressure and continues through the entire time cycle. Upon completion of the test, the caps shall be thoroughly cleaned and dried.

4.4.4.3 Hardness. The hardness of the cap body shall be determined by using a superficial hardness tester with a 15-T scale, 1/16 inch ball and a 15-kilogram load. A minimum of three readings shall be taken at various points within 1/4 inch of the edge enveloping the disc. This procedure shall be repeated at two additional locations on the cap body - on the skirt and approximately equidistent between top and bottom.

4.4.4.4 Cap push-out test. The cap shall be placed in a test fixture and subjected to the required force. The force shall be applied to a minimum diameter of 3.05 inches to the inside surface of the disc.

5. PREPARATION FOR DELIVERY

5.1 Packaging, level C. The components shall be packaged to provide adequate protection against corrosion, deterioration and damage from the supply source to the first receiving activity for immediate use or for storage under controlled humidity conditions.

5.2 Packing, level C. The components, packaged as specified in 5.1 shall be packed to insure carrier acceptance and safe delivery to the

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first domestic destination. Containers shall be in compliance with Uniform Freight Classification rules or the regulations of other common carriers applicable to the mode of transportation.

5.3 Marking. In addition to any special marking required by the contract or order, unit packages, intermediate packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Preproduction.

(1) Time allowed for supplier's submission of samples for Government test and evaluation after award of contract.
(2) Destination of samples when testing is performed by the Government.
(3) Packing, shipping and marking instructions when applicable.
(4) Time required for the Government to notify the supplier whether to proceed with production.

6.2 Batch. A batch is defined as that quantity of material which has been manufactured by some chemical process and subjected to some physical mixing operation intended to make the final product substantially uniform.

6.3 Cap leak test fixture. The test fixture will be Government furnished equipment (GFE).

Custodian:

Army - MU

Preparing activity:

Army - MU(EA)

Project No. 1330-A231

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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